

LECTURE 02 · REVIEW NOTES

Leveraged Investment Strategies

Margin accounts, leveraged ETFs, short selling and inverse funds, and how the mathematics of compounding changes the outcome of leveraged positions.

About these notes

These notes review the material of Lecture 02. Each concept is explained in full, and each numerical example is worked out step by step, with every intermediate figure shown. Interactive activities completed during class are summarized in general terms and marked as in-class activities.

1. Context: stock index performance, 2019-2026

Between January 2019 and September 2026 the three major U.S. stock indexes delivered very different results. The NASDAQ-100 (tracked by the QQQ ETF) gained **+260.5%**, the S&P 500 (SPY) gained **+182.4%**, and the Dow Jones Industrial Average (DIA) gained **+109.5%**. The difference is largely one of composition: the NASDAQ-100 is concentrated in large technology firms, while the Dow carries more industrial, energy and financial companies.

The same period also includes the 2022 bear market, in which the NASDAQ-100 lost roughly a third of its value. Higher long-run returns came with deeper drawdowns along the way. This raises the central question of the topic: what happens to an investor who applies borrowed money, or a 3× leveraged fund, to a path like this one? The answer turns out to involve much more than multiplying the return by three.

IN-CLASS ACTIVITY · Student presentation

A student team presented the evolution of the S&P 500, the Dow and the NASDAQ-100 over 2019-2026, comparing the 2019-21 bull market, the 2022 bear market and the full period. The discussion in class focused on why the indexes diverged and how drawdowns relate to long-run returns.

2. Background: ETFs, mutual funds and NAV

An investment company pools money from many investors and buys a portfolio. Its per-share value is the **net asset value** (NAV): what the fund owns, minus what it owes, divided by the number of shares outstanding.

$$\text{NAV} = (\text{Assets} - \text{Liabilities}) \div \text{Shares outstanding}$$

WORKED EXAMPLE 1**Computing NAV**

A fund holds \$500 million of stocks, owes \$5 million in accrued fees and other liabilities, and has 25 million shares outstanding.

Step	Calculation	Result
1	Net assets = Assets – Liabilities = \$500M – \$5M	\$495M
2	Divide by shares outstanding: \$495M ÷ 25M shares	NAV = \$19.80

Mutual funds and ETFs differ in how their shares are traded. A mutual fund share is bought from, or sold back to, the fund once a day at that day's closing NAV, so the transaction price is the NAV. An ETF share trades on an exchange throughout the day at a market price set by buyers and sellers, and the fund is not a party to most of those trades. That price can drift above NAV (a premium) or below it (a discount).

Authorized participants (APs), large dealers with a special agreement with the fund, keep that gap small. When the ETF trades above NAV, an AP delivers the underlying basket to the fund, receives newly created ETF shares worth NAV, and sells them on the exchange at the higher price; the added supply pushes the price back down. When the ETF trades below NAV, the AP does the reverse: it buys the cheap ETF shares and redeems them for the underlying basket, and the buying pressure lifts the price. For liquid funds such as SPY or QQQ the gap is usually 0.01–0.05%; in periods of market stress it can widen to 1–3%.

	ETF	Mutual fund
Trading	Intraday on an exchange	Once a day at NAV
Transaction price	Market price	End-of-day NAV
Minimum investment	One share	Often \$1,000+
Tax efficiency	High (in-kind redemptions)	Lower (capital-gain distributions)
Expense ratio	Very low (0.03–0.20%)	Higher (0.5–1.5%)
Can be sold short	Yes	No

This background matters because leveraged ETFs use exactly the same ETF structure. What they add is debt and derivatives, and a NAV that is reset to the target leverage every night.

3. Leveraged ETFs: the case of TQQQ

The ProShares UltraPro QQQ fund, ticker **TQQQ**, launched in February 2010, manages roughly \$22 billion (2025), and is the largest leveraged ETF in the world. Its objective is to

deliver **three times the daily return** of the NASDAQ-100. It charges an expense ratio of 0.88%, compared with 0.20% for QQQ, which tracks the same index without leverage.

The word *daily* is the most important part of that objective. TQQQ targets three times the return of each individual trading day, not three times the return over a week, a year or an investor's holding period. To meet that target, the fund rebalances at every market close so that its exposure again equals three times its NAV. As Section 7 shows, this nightly reset makes multi-day results differ, sometimes dramatically, from three times the index's multi-day return.

How the fund creates 3× exposure

TQQQ does not buy three dollars of Apple, Microsoft and NVIDIA for every dollar it receives from investors. The exposure is created synthetically. About 65% comes from **total return swaps**: over-the-counter contracts with large banks in which the bank pays the fund the NASDAQ-100's return and the fund pays the bank a floating financing rate (SOFR plus a spread). About 25% comes from **E-mini NASDAQ-100 futures**, exchange-traded contracts rolled every quarter. The remaining assets are held in **Treasury bills**, which serve as collateral for the swaps and earn short-term interest.

For this reason the holdings of TQQQ look nothing like those of QQQ. QQQ's largest positions are Apple (9.0%), Microsoft (8.2%), NVIDIA (8.0%) and the rest of the index, with the top 20 names making up about 68% of the fund. TQQQ's largest positions are six T-bills totaling roughly 85% of net assets, followed by swap contracts with Goldman Sachs, Morgan Stanley, Citibank, BNP Paribas, UBS and others, and a small futures position. It holds *no stocks at all*. The swaps appear at only a few percent of net assets because a swap is carried at its unrealized gain or loss, not at the size of the exposure it creates.

WORKED EXAMPLE 2

Holdings versus exposure

Assume TQQQ's NAV is \$100 per share and the NASDAQ-100 rises 1% in a day. The table traces the fund's position through the day and the rebalancing needed to restore 3× exposure for the next day.

Step	Calculation	Result
1	Required exposure at the start of the day = 3 × NAV = 3 × \$100	\$300 of index exposure
2	Source: the fund's own \$100 plus about \$200 of <i>notional</i> exposure financed through swaps and futures	Swap notional ≈ 200% of NAV
3	Index rises 1%: gain on exposure = 1% × \$300	+\$3.00
4	New NAV = \$100 + \$3 (ignoring one day of fees and financing)	\$103.00 (+3.0%)
5	Exposure after the move = \$300 × 1.01; target for the next day = 3 × \$103	\$303 held vs. \$309 needed
6	Rebalance at the close: add exposure equal to \$309 – \$303	Buy \$6 more exposure

Step 6 is the key mechanical point. After an *up* day a leveraged ETF must *add* exposure, and after a *down* day it must *cut* exposure. Leveraged funds are therefore forced buyers into strength and forced sellers into weakness, always near the close. The example also makes the cost of leverage explicit: the fund pays a financing rate on the borrowed \$200 of exposure every day, in addition to the 0.88% management fee.

Regulatory and systemic concerns

Several regulators have flagged these products. The IMF's *Global Financial Stability Report* (October 2023) described leverage in non-bank financial intermediation as a key vulnerability and noted that end-of-day rebalancing by leveraged ETFs mechanically amplifies directional moves. SEC staff (November 2022) warned retail investors that leveraged and inverse ETFs are complex products typically not suitable for long-term holding, and brokers must now assess suitability before recommending them. FINRA (2023) found that brokers often treated these funds like ordinary ETFs and rarely explained the daily compounding effect to clients.

The concerns fall into three channels. The first is **forced rebalancing**: as Example 2 shows, the leveraged-ETF sector as a whole buys after rallies and sells after declines, adding momentum to the market. The second is **counterparty concentration**: the fund's exposure sits with a small group of large banks, so stress at those banks reaches the fund immediately. The third is the **liquidity illusion**: TQQQ shares trade \$3–4 billion a day in normal markets, but the swaps underneath are OTC contracts that could gap in a crash. The liquidity of the ETF share is not the same as the liquidity of what the fund holds.

4. How debt amplifies returns

Leverage can be isolated from any ETF machinery by comparing two investors who each control a \$10,000 position in the same asset. The unleveraged investor contributes all \$10,000. The leveraged investor contributes \$5,000 and borrows the other \$5,000 at 6%

per year, a 2:1 ratio of assets to equity. Both hold exactly the same asset; only the financing differs. The lender's claim is **senior**: principal plus interest ($\$5,000 \times 6\% = \300) must be repaid whatever the asset does, and the investor keeps only what remains, the **residual**.

WORKED EXAMPLE 3

Same \$10,000 position, bull (+20%) and bear (−20%)

Step	Calculation	Result
1	Capital structure. Unleveraged: \$10,000 equity, \$0 debt. Leveraged: \$5,000 equity, \$5,000 debt at 6%	Both invest \$10,000
2 · Bull	Asset value after +20%: $\$10,000 \times 1.20$	\$12,000 (both)
3 · Bull	Leveraged investor repays the lender: $\$5,000 + 6\% \times \$5,000$	\$5,300
4 · Bull	Unleveraged ROE = $(\$12,000 - \$10,000) \div \$10,000$	+20%
	Leveraged: keeps $\$12,000 - \$5,300 = \$6,700$. Profit = $\$6,700 - \$5,000 = \$1,700$. ROE = $\$1,700 \div \$5,000$	+34%
2 · Bear	Asset value after −20%: $\$10,000 \times 0.80$	\$8,000 (both)
3 · Bear	The lender is still owed principal plus interest	\$5,300
4 · Bear	Unleveraged ROE = $(\$8,000 - \$10,000) \div \$10,000$	−20%
	Leveraged: keeps $\$8,000 - \$5,300 = \$2,700$. Loss = $\$2,700 - \$5,000 = -\$2,300$. ROE = $-\$2,300 \div \$5,000$	−46%

The asset moved the same $\pm 20\%$ for both investors, yet the leveraged investor earned +34% or lost 46%. The effect is also *asymmetric*: the upside is amplified by 14 percentage points and the downside by 26, because the \$300 of interest is owed in both states. The same result follows from a general formula for any leverage ratio:

$$\text{ROE} = r + (D \div E) \times (r - i)$$

where r is the asset's return, i the borrowing rate and $D \div E$ the debt-to-equity ratio (here $\$5,000 \div \$5,000 = 1$). Bull: $20\% + 1 \times (20\% - 6\%) = 34\%$. Bear: $-20\% + 1 \times (-20\% - 6\%) = -46\%$. The spread ($r - i$) determines whether leverage helps: it does only when the asset earns more than the cost of borrowing. The break-even is $r = i = 6\%$; below that, borrowing leaves the leveraged investor worse off.

5. Buying on margin and the margin call

A margin account is the mechanism through which an individual investor borrows from a broker to buy securities. Two percentages govern it. The **initial margin** is the minimum share of the purchase that must be funded with the investor's own money; under the Federal Reserve's Regulation T it is 50% for stocks. The **maintenance margin** is the minimum equity that must be kept afterwards; the example below uses 30%, and brokers often set their own higher levels. At any moment the margin in the account is:

$$\text{Margin} = \text{Equity} \div \text{Market value} = (P \times Q - \text{Loan}) \div (P \times Q)$$

The loan is fixed in dollars, while the value of the shares moves with the price. Every price decline therefore comes entirely out of the investor's equity, and the margin percentage falls faster than the price does.

WORKED EXAMPLE 4

Buying 100 shares at \$10 with a \$500 loan

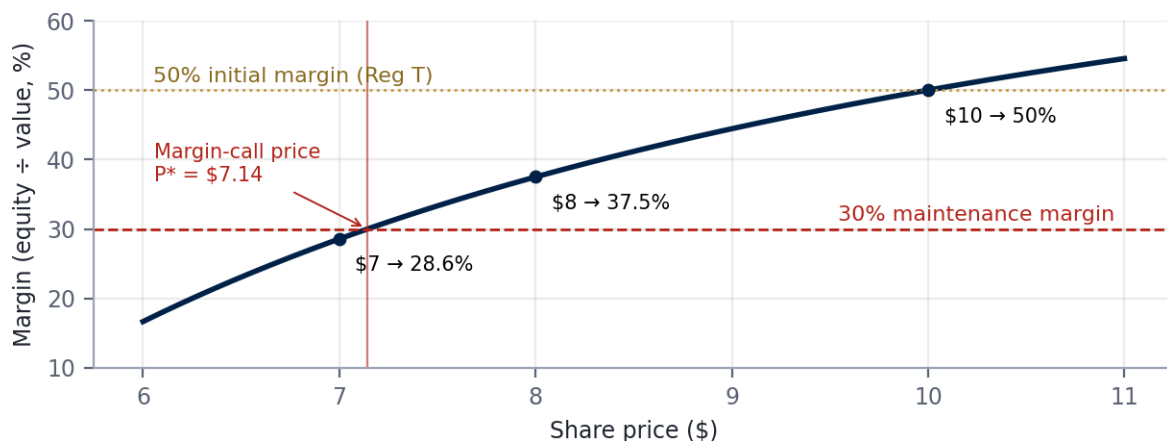
Step	Calculation	Result
1 • Open	Position value = $100 \times \$10$	\$1,000
	Equity = $\$1,000 - \500 loan	\$500
	Margin = $\$500 \div \$1,000$ (Reg T minimum is 50%)	50%, allowed
2 • \$8	Position value = $100 \times \$8$; loan unchanged at \$500	\$800
	Equity = $\$800 - \500	\$300
	Margin = $\$300 \div \800	37.5%, above 30%
3 • \$7	Position value = $100 \times \$7$	\$700
	Equity = $\$700 - \500	\$200
	Margin = $\$200 \div \700	28.6%: MARGIN CALL

Between the purchase and a price of \$7, the share price fell 30% but the investor's equity fell from \$500 to \$200, a 60% loss: the same 2:1 leverage as in Example 3. With margin, however, losses do not simply accumulate. Once the margin falls below the maintenance level, the broker issues a **margin call**, and the investor must add cash or securities immediately or the broker may sell the shares without further notice.

Finding the margin-call price

Rather than testing prices one at a time, the exact price P^* at which the margin equals the maintenance requirement m can be found by setting the margin formula equal to m and solving for P :

Step	Algebra	Result
1	Start from the definition and set it equal to m	$(P \times Q - L) \div (P \times Q) = m$
2	Multiply both sides by $P \times Q$	$P \times Q - L = m \times P \times Q$
3	Collect the P terms on one side	$P \times Q \times (1 - m) = L$
4	Divide by $Q \times (1 - m)$	$P^* = L \div [Q \times (1 - m)]$
5	Substitute $L = \$500$, $Q = 100$, $m = 0.30$: $\$500 \div (100 \times 0.70) = \$500 \div 70$	$P^* = \$7.14$
6	Check: value = \$714.29; equity = \$714.29 – \$500 = \$214.29; $\$214.29 \div \714.29	= 30% exactly



Margin percentage as the share price changes, for 100 shares bought with a \$500 loan. The curve falls faster as the price drops because the loan does not shrink.

Meeting the call. At \$7 the equity is \$200, while 30% of \$700 is \$210. A cash deposit of **\$10**, which reduces the loan to \$490, restores the margin to exactly 30% ($\$210 \div \700). Many brokers require the account to be brought back to the initial 50%, which needs equity of $0.50 \times \$700 = \350 , a deposit of **\$150**. If the investor cannot pay, the broker sells shares at the worst possible moment, after the price has already fallen. This is why margin calls are described as *forced selling*, and why leverage that seems manageable in calm markets can wipe out an account in a crash.

6. Short selling

Short selling allows an investor to profit when a price falls. The process has four steps. First, the investor **borrow**s shares through the broker, which locates them in other margin accounts or with institutional lenders; the investor pays a borrow fee, typically around 0.3% a year for easy-to-borrow stocks and 20% or more for hard-to-borrow ones. Second, the investor **sell**s the borrowed shares at the market price; the proceeds are credited to the account but held as collateral. The investor now owes shares, not dollars. Third, the position stays open: a price decline produces a profit and a price increase a loss. Fourth, the investor **cover**s by buying the shares back and returning them to the

lender, either voluntarily or because a margin call forces it.

$$\text{Profit} = (\text{Sale price} - \text{Cover price}) \times \text{Shares} - \text{Borrow fees} - \text{Dividends paid}$$

WORKED EXAMPLE 5

Shorting 100 shares of TSLA at \$300 for 60 days

Step	Calculation	Result
1	Proceeds from the short sale = $100 \times \$300$ (held as collateral)	\$30,000
2	Borrow fee at 2% per year for 60 days = $\$30,000 \times 0.02 \times 60 \div 365$	\$98.63 ≈ \$99
3A	Scenario A, cover at \$240: gross gain = $(\$300 - \$240) \times 100$	+\$6,000
	Net profit = $\$6,000 - \99	+\$5,901
3B	Scenario B, cover at \$380: gross loss = $(\$300 - \$380) \times 100$	-\$8,000
	Net result = $-\$8,000 - \99	-\$8,099
4	Break-even cover price: the decline must cover the fee, $\$300 - \$99 \div 100$	≈ \$299.01

The example illustrates the basic **asymmetry of shorting**. The maximum gain on a short position is 100% of the sale price, reached if the stock goes to zero. There is no ceiling on the loss, because there is no ceiling on how high a stock can rise; in Scenario B the loss keeps growing for as long as the price climbs. Short sellers also face risks that buyers do not. In a **short squeeze** (GameStop, January 2021), rising prices force short sellers to cover, and their buying pushes the price higher still. The lender can **recall** the shares at any time. And any **dividend** paid while the position is open must be paid by the short seller to the lender.

7. Inverse ETFs and volatility decay

SQQQ, the ProShares UltraPro Short QQQ fund, targets **-3× the daily return** of the NASDAQ-100. It allows an investor to take a position against the index without borrowing shares, and the maximum loss is the amount invested. Its long-run expected return is negative, however: the stock market drifts upward over time, and the daily reset adds **volatility decay**, the most important idea of this topic. SQQQ is a tool for short, well-timed hedges. In 2022 it gained about +79% while QQQ fell 33%, but capturing that gain required getting both the entry and the exit right.

WORKED EXAMPLE 6

Two volatile days that go nowhere

Consider \$1,000 invested in QQQ, in a 3× fund and in a −3× fund. On Day 1 the index rises 10%; on Day 2 it falls 10%.

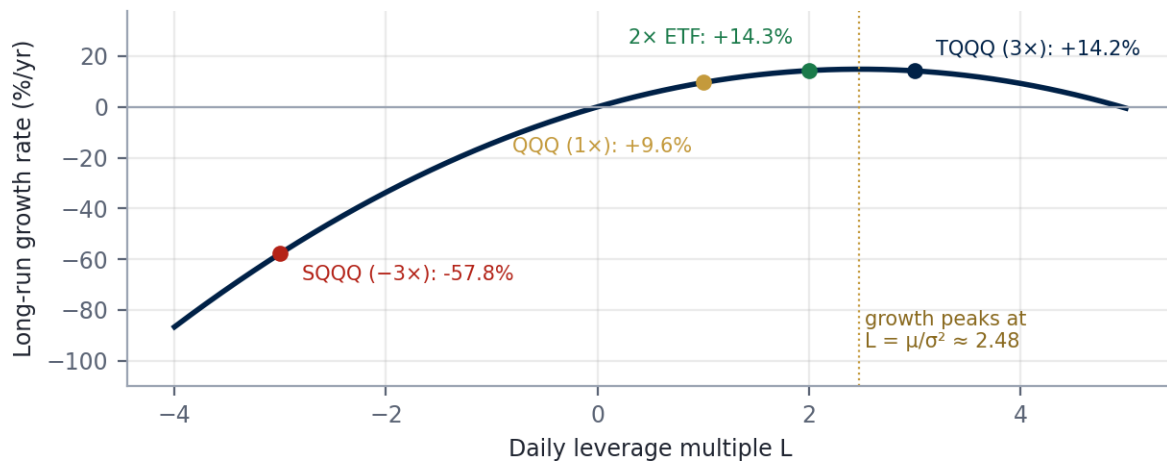
	QQQ (1×)	3× fund (TQQQ)	−3× fund (SQQQ)
Day 1 return	+10%	$3 \times 10\% = +30\%$	$-3 \times 10\% = -30\%$
Value after Day 1	$\$1,000 \times 1.10 = \$1,100$	$\$1,000 \times 1.30 = \$1,300$	$\$1,000 \times 0.70 = \700
Day 2 return	−10%	$3 \times (-10\%) = -30\%$	$-3 \times (-10\%) = +30\%$
Value after Day 2	$\$1,100 \times 0.90 = \990	$\$1,300 \times 0.70 = \910	$\$700 \times 1.30 = \910
Two-day return	$(1.10)(0.90) - 1 = -1.0\%$	$(1.30)(0.70) - 1 = -9.0\%$	$(0.70)(1.30) - 1 = -9.0\%$
Naive '3 × index'	n/a	$3 \times (-1\%) = -3\%$	$-3 \times (-1\%) = +3\%$

The index lost only 1%, yet the 3× fund lost 9%, three times more than the naive −3%. More strikingly, the *inverse* fund, which might be expected to gain when the index falls, *also* lost 9%. Both leveraged funds lost because a gain followed by an equal percentage loss always ends below the starting point, and leverage magnifies both legs: $(1 + x)(1 - x) = 1 - x^2$. For the index this is $1 - 0.10^2 = 0.99$; for either $\pm 3\times$ fund it is $1 - 0.30^2 = 0.91$. The loss grows with the *square* of the daily move, which is why decay rises so quickly with leverage and volatility.

IN-CLASS ACTIVITY · Volatility decay simulator

An interactive simulator was used in class to compare 1×, 2×, 3× and −3× daily-reset funds over a year of simulated trading, with adjustable volatility, drift and number of days. The general lessons: the drag grows with the *square* of leverage and with volatility, so more leverage does not necessarily mean more long-run growth, and an inverse fund loses both from the market's upward drift and from the same volatility drag that affects leveraged long funds.

A standard approximation summarizes the effect: long-run growth $\approx L \times \mu - \frac{1}{2} \times L^2 \times \sigma^2$, where L is the daily leverage, μ the index drift and σ its volatility. The figure below shows its general shape.



Approximate long-run growth rate as a function of daily leverage, for an index with 12% drift and 22% volatility, before fees and financing costs.

Recovery math: why drawdowns hurt so much

After a loss of x , a gain of $x \div (1 - x)$ is needed just to return to the starting value. A 33% loss (QQQ in 2022) requires $0.33 \div 0.67 = +49\%$. A 50% loss requires **+100%**. A 79% loss, roughly TQQQ's decline in 2022, requires $0.79 \div 0.21 = +376\%$. Leverage deepens drawdowns, and the required recovery rises much faster than the drawdown itself.

8. Key takeaways

Leverage is a multiplier in both directions. Borrowing to invest raises the return on equity when the asset earns more than the borrowing rate, and deepens the loss when it does not. The ROE formula $r + (D/E)(r - i)$ applies whether the leverage comes from a margin loan or from inside a 3× ETF.

A daily reset changes everything. TQQQ delivers 3× the *daily* return, not 3× the return over time. Because volatility drag grows with the square of leverage, a 3× fund can trail three times the index return even in a rising market, and lose heavily in a volatile, sideways market.

Margin calls turn investors into forced sellers. The loan is fixed and the collateral is not, so a price decline erodes equity at a leveraged rate. The margin-call price $P^* = L \div [Q(1 - m)]$ identifies in advance where that forced selling begins.

Short selling has asymmetric risk. The gain is capped at 100% and the loss is unlimited, and short squeezes, share recalls and dividends owed add risks that buyers never face.

Leveraged products matter for the whole system. The IMF, SEC and FINRA treat leveraged ETFs as a source of market-wide risk because their end-of-day rebalancing feeds back into prices.

Holdings are not the same as exposure. TQQQ holds T-bills and swap contracts, not a single share of Apple or NVIDIA, yet it carries three times the NASDAQ-100's daily risk. Analyzing a portfolio's risk requires looking through its holdings to its economic exposure.

Figures on index performance, fund characteristics and regulatory statements are those presented in Lecture 02. The growth-rate approximation in Section 7 is a standard continuous-time result that ignores fees, financing costs and path details.